1. What is AI Ethics?

Explanation

- “Ethics” means a set of moral principles about what is right or wrong. DataCamp+1
- AI ethics refers to the field that studies how we should design, build, deploy and use AI systems so they abide by human values, minimise harm and maximise benefit. IBM+2AI Business+2
- Another way: It’s applying moral frameworks into technological context — for AI this means privacy, fairness, accountability, transparency, safety, etc. DataCamp+1

Why do we need to teach it?

- AI systems now influence many parts of life: health care, finance, hiring, law enforcement, etc. Ethical issues matter when decisions affect people. PMC+1
- Without ethics: AI could replicate or amplify existing biases, invade privacy, reduce trust, cause harm. DataCamp+1
- Ethical design helps build trustworthy AI, which in turn helps adoption, societal benefit. Forbes+1

What happens if we don’t have AI ethics?

- Example: Biased algorithms might unfairly disadvantage certain groups. ISACA+1
- Loss of trust: People might stop using AI systems if they feel they are unfair or opaque. University of the People
- Societal harm: For example, unfair decisions in hiring/credit/law-enforcement. annenberg.usc.edu
- Innovation risk: If AI causes harm, regulation may become stricter or public backlash may slow progress.

---

2. Key Concepts to Cover

A. Bias & Detection (Bias in AI)

What is bias in AI?

- "Bias" in the AI context means that an AI system favours or disfavors certain individuals/groups in a way that is unfair. PMC+1
- This can arise from skewed training data, algorithmic design, or even how decisions are used. CloudThat+1

Why detection is important

- Because if bias goes unchecked, decisions made by AI may systematically disadvantage minorities or protected groups. Ethics Unwrapped+1
- It affects trust, legal compliance, reputation.

Simple example for students

- Suppose an AI model for loan approval is trained mostly on male applicants and less on females → it may learn "male → good risk" more than female → unfair.
- Or a facial-recognition tool trained mostly on light-skinned people fails for dark-skinned people. (A case discussed in literature) Medium

B. Fairness

What is fairness in AI?

- Fairness means that an AI system gives people impartial, just, non-discriminatory treatment. SS&C Blue Prism+1
- Note: Fairness is tricky because "everyone is treated the same" may not always be fair (because groups start at different positions). Stanford HAI

Why fairness matters

- To protect people's rights (e.g., not discriminated by gender, race, etc.). clanx.ai

- To ensure we can trust AI systems and that they serve society, not harm. Built In

Simple classroom discussion

- Ask: If an AI chatbot gives faster responses to premium members than free members, is that fair? What if free-members are mostly from a historically disadvantaged group?
- Show that fairness also means checking data and process: Are all groups represented? Are decisions transparent?

C. AI Regulation

What is AI regulation?

- Regulation of AI means laws, guidelines and policies by governments/regulators to oversee development, deployment and use of AI. Orrick
- It aims to make sure AI systems are safe, reliable, transparent, respect rights, and minimise risk. Wikipedia

Why regulation is necessary

- Because technology moves fast, and ethical practices alone may not guarantee protection of rights or safe outcomes.
- Regulation helps build a baseline of trust, sets expectations, holds actors accountable.

Example: The EU AI Act

- The EU has introduced a regulation called AI Act which classifies AI systems by risk (e.g., unacceptable risk = banned; high risk = specific obligations). European Parliament+1
- This shows one model of how regulation can apply to AI.

---

3. Case Studies / Real-world Illustrations

- The website of AI Ethicist lists many AI ethics case-studies and incidents you can use for discussion. AI Ethicist
- Example case: Hiring algorithms that seemed neutral but ended up preferring male applicants because training data had mostly men. Ethics Unwrapped
- Example: Bias in algorithmic decision-making in healthcare or social media, leading to unfair outcomes for certain groups. PMC+1

You can pick 1-2 cases for your students to read/discuss:

- What went wrong?
- Which ethical principle was violated (fairness, transparency, accountability, etc.)?
- How might it have been prevented?

---

4. Linking the Topics Together

- Bias detection → helps make AI fairer by identifying unfair outcomes.
- Fairness → a goal for AI systems; you measure & build towards it.
- Regulation → helps enforce fairness and bias-mitigation at a larger scale and ensures accountability.
- Without all three, we risk building AI that works technically but harms socially or legally.

---

5. Lecture Flow Suggestion (for ~1–1.5 hours)

1. Introduction: What is AI ethics (definition, need)
2. Why it's important (with examples of harms if missing)
3. Bias in AI: definition, sources, simple example, group exercise

4. Fairness: What it means, challenges (e.g., treating everyone equally vs equitably), discussion
5. Regulation: What it is, why needed, example of EU AI Act, discussion on how this affects India/other countries
6. Case study review: Break students into pairs to pick a case, answer guided questions
7. Wrap-up: Key take-aways, open questions for students to think of research topics or short projects (fits your “small research project” aim)

---

6. Suggestions for Student Activities / Mini-Research Projects

- Each student-pair picks an AI application (e.g., loan approval, hiring, medical diagnosis, facial recognition) and investigates:
    - What data is used? Could there be bias?
    - What fairness issues might arise?
    - Are there regulations in India for that application?
    - How could the system be made more ethical (bias mitigation, fairness checks, transparency)?
- Write a short report: literature review (what is known), gap (what’s missing), methodology (how you’d check bias/fairness), expected result/conclusion.
- Present a mini-case in class.

---

7. Key Take-away Points for Students

- Ethics in AI is **not optional** — it is essential for social trust, safety, legal compliance and responsible innovation.

- Bias is pervasive and can sneak into AI systems via data, design, or usage; detecting and mitigating it is part of fairness.
- Fairness is a complicated but essential goal — there is no one "perfect" way, but striving for equitable outcomes is important.
- Regulation is catching up globally: it helps set common standards, enforce rights, and guide responsible AI.
- As future professionals (BCA students), being aware of these issues will help them not just technically but ethically design or use AI systems.

1. Bias Detection

What exactly is bias in AI?

- "Bias" means that an AI system (algorithm/model) treats people differently in a way that's unfair, often because the data or the design reflected past unequal patterns. docs.mend.io+2Google for Developers+2
- For example: If an AI hiring tool is trained mostly on male candidates, it might favour men even when women are equally qualified. docs.mend.io+1
- It can show up as: some groups having lower approval rates, higher error rates, excluded altogether, or being mis-classified more often.

How do we detect bias?

Here are some steps and methods:

1. **Define the sensitive attributes**: Decide which groups you care about (e.g., gender, race, age, income level).
2. **Collect data**: You need data with those attributes plus the model's predictions and true outcomes.

3. **Compare outcomes across groups**: Look at how the model behaves for different groups. Example: Does Group A get positive outcomes much less often than Group B?

4. **Use bias detection tools or metrics**: There are toolkits and methods. For example, the document "Bias (Mend.io)" explains structured testing of bias: generate test-cases, simulate different groups, check the differences. docs.mend.io

5. **Visualise and interpret**: It helps to have charts/tables showing differences (for example approval rate by group).

6. **Investigate root-causes**: If bias is found, try to understand why — is it due to skewed data, historical discrimination, algorithmic choices, threshold settings?

7. **Monitor continuously**: Bias can creep in over time when data changes, so monitoring after deployment is important. CSIRO Research

Why detection matters

- Because if bias is not detected, the system might produce unfair decisions and harm individuals or groups.

- It undermines trust and can lead to legal/regulatory issues.

- For students: You can ask: *If you deploy an AI for loan approvals, how would you check if it is biased?*

---

2. Fairness Metrics

What are fairness metrics?

- Fairness metrics are **quantitative ways** to measure how fair a model's outcomes are across different groups. They help turn "Is it fair?" into "How fair is it?" enz.ai

- There are many different metrics, and which one you choose depends on context (what application the AI is used for) and what kinds of fairness matter in that context. alan-turing-institute.github.io+1

Key fairness metrics (with simple explanation)

Here are some common ones:

- **Demographic Parity (or Statistical Parity)**: The proportion of people getting a positive outcome (e.g., loan approved) should be the same for all groups. GeeksforGeeks+1
  - Example: If 70% of men get loans, then women should also get ~70% if we want demographic parity.
  - But one challenge: The true underlying risk might differ between groups, so blindly enforcing this can reduce accuracy. GeeksforGeeks
- **Equalized Odds**: For each group, the true positive rate (TPR) *and* the false positive rate (FPR) should be the same. GeeksforGeeks+1
  - Example: If 80% of qualified men are approved (TPR) and 10% of unqualified men get approved (FPR), then for women you also want ~80% TPR and ~10% FPR.
- **Equality of Opportunity**: A simpler version: make sure the true positive rates (for the "positive" class) are equal across groups. (Concerned more with fairness among those who *should* get positive outcomes). alan-turing-institute.github.io
- **Disparate Impact**: Often expressed as a ratio: (Positive rate for un-privileged group) ÷ (Positive rate for privileged group). If this ratio drops below some threshold (often 0.8) it's a red flag. aialchemyhub.in
- **Calibration**: Predictions should correspond to actual outcomes equally across groups. For example, if the model predicts "80% chance of success" for people in Group A and Group B, in both groups about 80% should in fact succeed. enz.ai

How to measure these – simple steps

1. Compute the model's output predictions & true labels for each group.
2. Compute the relevant rates: positive outcome rate, TPR, FPR, etc.
3. Calculate the metric (difference or ratio). Example: difference between TPRs of groups; or ratio of positive rates.
4. Decide threshold or acceptable range. For instance, many fairness standards say if ratio < 0.8 (the "4/5th rule"), it might be unfair. TEC

5. Interpret and decide action: If the metric indicates unfairness, adjust model/data (e.g., re-weighting, change threshold, add more data for underrepresented group). aialchemyhub.in+1

Some caveats / things to tell students

- No one metric fits *all* situations. Sometimes you cannot satisfy all metrics at once (there are trade-offs). alan-turing-institute.github.io
- Fairness vs Accuracy: Enforcing strict fairness may reduce accuracy for some groups or overall. GeeksforGeeks
- Context matters: What is "fair" in one domain may differ in another (loan approval vs medical diagnosis vs hiring).
- Fairness for groups vs individuals: Most metrics are group-based (comparing groups). Individual fairness (treating similar individuals similarly) is harder but important. CSIRO Research

---

3. AI Regulation

What is AI regulation exactly?

- AI regulation refers to **laws, rules, guidelines, standards** created by governments or international bodies to oversee how AI systems are developed, deployed, and used. Encyclopedia Britannica+1
- The purpose is to ensure rights, safety, fairness, transparency, accountability in AI systems — especially since AI has potential for large societal impact (positive and negative). Nemko Digital+1

Examples of AI regulation around the world

- The Artificial Intelligence Act (EU): A regulation by the European Union which entered into force on 1 August 2024. It sets out risk-based obligations for AI systems. Wikipedia+1

- China's "Interim Measures for the Management of Generative AI Services": Rules introduced in 2023 governing public-facing generative AI services. Wikipedia+1

- Global overviews: Many countries have AI strategies, voluntary guidelines, sector-specific laws. Nemko Digital+1

What do these regulations cover / require?

Some of the things regulations typically require:

- **Risk-classification** of AI systems: e.g., some AI uses are "high risk" (healthcare, law enforcement) and need stronger controls. Nemko Digital

- **Transparency**: AI systems must be explainable, users must be informed when they are interacting with AI, etc.

- **Data governance and quality**: Ensuring the training data is appropriate, not discriminatory, and traceable.

- **Human oversight**: Ensuring humans can oversee or override AI decisions when necessary.

- **Accountability / liability**: Who is responsible if the AI causes harm?

- **Monitoring and audit**: After deployment, systems must be monitored for performance, fairness, safety, and updates.

- **Prohibitions on certain uses**: Some practices might be banned (e.g., social scoring of individuals, real-time biometric identification in public spaces, depending on region) under EU rules. Reddit

How to study: What to measure or check for regulation compliance

- **Categorise your system**: Is it a low-risk, medium-risk or high-risk AI use? (Based on domain).

- **Documentation**: Ensure you have documentation of training data, model design, fairness/bias testing, performance.

- **Fairness/bias audits**: Use fairness metrics (see above) and bias detection methods to test your system.

- **Transparency check**: Are decisions explainable? Are users informed? Can human override happen?
- **Post-deployment monitoring**: Are you tracking performance, fairness metrics, error rates over time?
- **Regulatory updates**: Are you aware of local/national regulations (India, EU, etc) and ready to comply?

Why regulation matters (and some pros & cons)

Pros

- Improves trust in AI systems among users and society. Encyclopedia Britannica
- Helps prevent harm (bias, discrimination, privacy breaches).
- Provides clear rules for companies, reduces legal uncertainty.

Cons / challenges

- Over-regulation could slow innovation or raise costs (especially for small/start-ups). Encyclopedia Britannica
- Regulations might become outdated as technology advances quickly.
- Global fragmentation: Different regions have different rules → companies working internationally face complexity. arXiv

---

Bringing it all together

- Bias detection → means finding when an AI model is unfairly treating groups differently.
- Fairness metrics → are the *tools* for measuring how fair the treatment is (quantitative checks).
- AI regulation → is the *external framework* (laws, rules) that enforce or encourage fair, safe, transparent AI use.